

Custom connectors in TRENTOS

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Status

- Used to implement threads in a demo_hello https://bitbucket.hensoldt-cyber.systems/projects/SS/repos/demo_hello_world/browse?at=refs%2Fheads%2FSEOS-thtst
- Used for a network stack notification https://bitbucket.hensoldt-cyber.systems/projects/SS/repos/test_network_api/browse?at=refs%2Fheads%2FSEOS-666-connector

Links

- <https://docs.sel4.systems/projects/camkes/internals.html#template-context>
- <https://github.com/sel4/camkes-tool/blob/master/docs/index.md#templating>

Declaration

A connector's files can be located anywhere we like, the directory just needs to be added to the CAMkES Template path in the cmake file.

Declaring connector in cmake

```
CAMkESAddTemplatesPath(<directory_to_look_for_connector>)

DeclareCAMkESConnector(
  <connector_name>
  FROM
  connector-from-template.c
  FROM_HEADER
  connector-from-template.h
  TO
  connector-to-template.c
  TO_HEADER
  connector-to-template.h
)
```

In the camkes file we declare the connector (with the same name used in cmake's DeclareCAMkESConnector call). If we don't want to have a thread on one of the sides, we declare it as Procedures with 0 threads

Declaring connector in camkes

```
connector <connector_name> {
  from Procedures;
  to Procedure;
  attribute int id = 3;
}

connector <connector_name> {
  from Procedures with 0 threads;
  to Procedure;
  attribute int id = 3;
}
```

Parsing attributes

Configuration values can be parsed from the camkes config using a construct like this:

```
/*- set id = configuration[c.instance.name].get("%s_id" % c.interface.name, 88) -*/
```

The value can be set in the camkes configuration block by

```
<component>.<interface>_id = 3;
```

For a comprehensive example, check out the connector defined in this branch

- Used for a network stack notification https://bitbucket.hensoldt-cyber.systems/projects/SS/repos/test_network_api/browse?at=refs%2Fheads%2FSEOS-666-connector